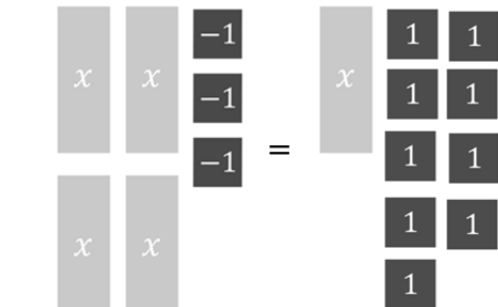
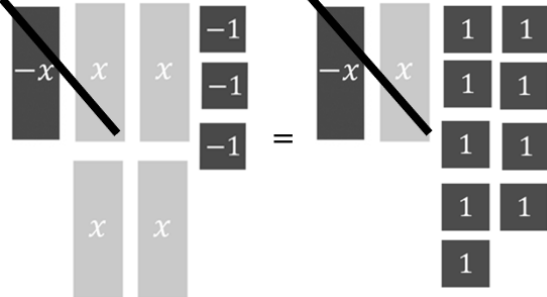
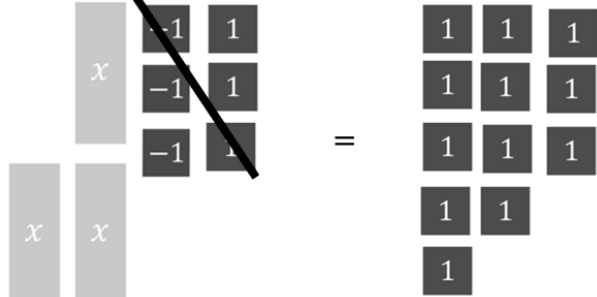
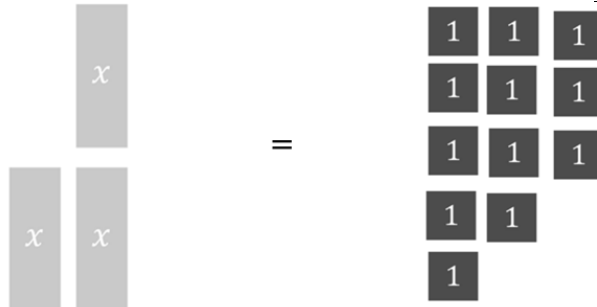


Grade 8  
Unit 8  
Lesson 3 Practice

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Use algebra tiles to solve the equation  $4x - 3 = x + 9$ .

1. Complete the table to describe the process for solving the equation.

| Algebra Tile Model                                                                  | Description                                                                    | Algebraic Equation                         |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------|
|    | Given Equation                                                                 | $4x - 3 = x + 9$                           |
|   | Add <u><math>-x</math></u> to both sides of the equation to create zero pairs. | $4x - 3 = x + 9$<br>$-x \quad -x$          |
|  | Add <u>3</u> to both sides of the equation to create zero pairs.               | $3x - 3 = 9$<br>$+3 \quad +3$<br>$3x = 12$ |
|  | Divide both sides of the equation into <u>3</u> equal groups.                  | $3x = 12$<br>$x = 4$                       |

2. Based on the work done in the table, what is the solution to  $4x - 3 = x + 9$ ?  
 $x = 4$

3. Solve each equation by following the steps described in the table.

| Description                                                                 | Algebraic Equation                    |
|-----------------------------------------------------------------------------|---------------------------------------|
| Begin with the given equation.                                              | $5(1.2m - 0.3) = -2m + 1.7$           |
| Apply the distributive property.                                            | $6m - 1.5 = -2m + 1.7$                |
| Eliminate the constant term from one side of the equation using zero pairs. | $+1.5 \quad +1.5$<br>$6m = -2m + 3.2$ |
| Eliminate the variable term from one side of the equation using zero pairs. | $+2m \quad +2m$<br>$8m = 3.2$         |
| Isolate the variable.                                                       | $m = 0.4$                             |

4. Check your answer for question #3 using substitution.

$$5(1.2 \cdot 0.4 - 0.3) = -2 \cdot 0.4 + 1.7$$

$$0.9 = -0.8 + 1.7$$

$$0.9 = 0.9 \checkmark$$

5. Solve the equation  $8 - k = 3k + 20$ .

$$+k \quad +k$$

$$8 = 4k + 20$$

$$-20 \quad -20$$

$$-12 = 4k$$

$$\frac{4k}{4} = -\frac{12}{4}$$

$$k = -3$$

6. Check your answer using substitution.

$$8 - (-3) = 3(-3) + 20$$

$$11 = 11 \checkmark$$

7. Solve the equation  $4(s - 5) = 9s + 25$ .

$$4s - 20 = 9s + 25$$

$$-4s \quad -4s$$

$$-20 = 5s + 25$$

$$-25 \quad -25$$

$$-45 = 5s$$

$$-9 = s$$

8. Check your answer using substitution.

$$4(-9 - 5) = 9(-9) + 25$$

$$-56 = -56 \checkmark$$

9. Solve the equation  $3(1 - 3p) = 6(p - 7)$ .

$$3 - 9p = 6p - 42$$

$$+9p \quad +9p$$

$$3 = 15p - 42$$

$$+42 \quad +42$$

$$\frac{45}{15} = \frac{15p}{15}$$

$$p = 3$$

10. Check your answer using substitution

$$3(1 - 3 \cdot 3) = 6(3 - 7)$$

$$3 \cdot -8 = -24$$

$$-24 = -24 \checkmark$$